

Roman Knyazhitskiy

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Summary

Research Engineer at Mirai Labs, enjoying efficient ML, learning theory and robotics. My side interests include abstract interpretation, philosophy of science and AI Safety.

Education

MPhil in MLMI (Machine Learning), University of Cambridge 10/2025 – 09/2026

BSc Computer Science and Engineering + Honours, TU Delft 09/2022 – 07/2025

Work Experience

Research Engineer, Mirai Labs 01/2026 – now

- Lead post-training quantization, achieving lower held-out KL than Unsloth at matched bpw.
- Responsible for the full evaluation suite; assembled our main training data mixture.
- Contributed across the research stack.

Research Associate, TU Delft 03/2023 – 08/2025

- Developed a transformer-based Prior-Data Fitted Network (PFN) for hyperparameter optimization, achieving 15–20% improvement over the current SOTA.

Student Team ML Lead, Delft Mercurians 05/2023 – 09/2025

- Led machine learning for an autonomous robot soccer team (RoboCup Small Size League).
- Deployed a real-time Model Predictive Control system on hardware (15 ms latency), interfacing Python with Rust.

Projects

- **Continuous-time Differentiable Simulator (2023)**: Implementation of a small BRAX-like physics engine in JAX, with simple solvers and better multi-body support.
- **Stack Associated Beam Tracing (2022)**: Likely saymptotically fastest octree tracing/rendering algorithm at the time.

Honours and Awards

- **1st Place**, Bunq Hackathon 6 (2025); €30,000 prize
- **2nd Place & Special Prize**, Epoch AI Hackathon (2024)
- **Best Software Solution**, RoboCup World Championships, Sydney (2019)

Other

- Contributed to **jaxtyping**, **Equinox** and **beartype** stack, attempting to improve runtime typechecking for users of IPython notebooks.

Publications

- [1] R. Knyazhitskiy, R. Mathieu, and A. Chakhvadze. Approximate Vocabulary Projection via Reduced Regression. Submitted to AAI, 2026.
- [2] Y. Oda, R. Mathieu, R. Knyazhitskiy, and A. Chakhvadze. **Trees from Marginals: Autoregressive Drafting with Factorized Priors**. arXiv preprint, 2026.
- [3] R. Knyazhitskiy and A. G. Di Francesco. **Early-Exit Graph Neural Networks for Link Prediction**. Accepted at LoG@Pisa, 2026.
- [4] R. Knyazhitskiy and P. R. Van der Vaart. **A Simple Scaling Model for Bootstrapped DQN**. Accepted at EWRL@Lille, 2026.
- [5] J. Luijmes, A. Gielisse, R. Knyazhitskiy, and J. van Gemert. **ARC: Anchored Representation Clouds for High-Resolution INR Classification**. Accepted at ICLR Workshop on Weight Space Learning, 2025.